

Polystyrene Hydrogenolysis to High-Quality Lubricants Using Ni/SiO₂

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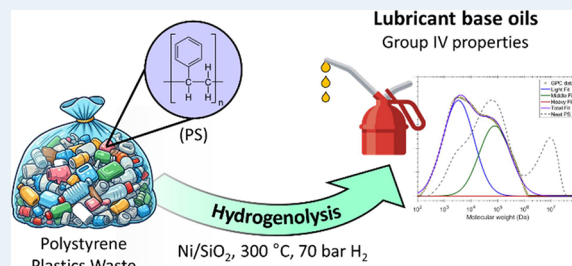
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ABSTRACT: Pyrolytic and light-activated oxidation processes are leading technologies for utilizing polystyrene (PS) wastes. These approaches exhibit poor selectivities, use complex reactors, and require solvents. Hydrogenolysis is effective for deconstructing polyolefins, but its application to PS feedstocks has been limited. Herein, we demonstrate Ni/SiO₂ catalysts to facilitate PS ($M_w \approx 97$ kDa) hydrogenolysis to produce lubricant base oils possessing group IV properties, achieving maximum yields of 70% within 6 h at 300 °C and 70 bar of H₂. Gas, liquid, and oil product yields are stable across reaction conditions, whereas hydrogenation of the PS aromaticity and reduction of the molecular weight benefit from higher temperatures and H₂ pressures. Time-dependent experiments underscore the importance of elevated H₂ pressure, revealing that PS hydrogenolysis occurs sequentially, with aromatic ring hydrogenation preceding degradation of the C–C backbone. Kinetic measurements with 1,2-diphenylethane as a probe molecule demonstrate that ring hydrogenation is 3 orders of magnitude faster than internal C–C bond cleavage over Ni/SiO₂. Ni/SiO₂ proves to be effective in the hydrogenolysis of heavier PS polymers and rigid commercial PS products. Conversely, flexibility and foam PS feeds result in Ni/SiO₂ deactivation, attributed to performance additives. Unlike polyolefins, the process produces very little methane and other light hydrocarbons. These findings expand the applicability of hydrogenolysis to PS feedstocks, offering a versatile solution and broadening the range of high-value products from PS to include lubricant base oils.

KEYWORDS: plastic waste, hydrogenolysis, nickel, Earth-abundant metals, lubricants



1. INTRODUCTION

The massive scale of plastic utilization and unsustainable disposal of plastic waste have resulted in extensive strain on global ecosystems^{1–4} and significant consumption of the global carbon budget.^{5–8} A paradigm shift in the plastic waste industry is urgently needed to drive the transition from today's linear (take, make, and waste) economy⁹ to a future circular one. Mechanical recycling is a mature, cost-effective technology^{8,10,11} for polyethylene terephthalate and high-density polyethylene. Other thermoplastics are primarily diverted to landfills or incinerators.^{5,6}

Chemical recycling is promising for the utilization of these wasted thermoplastics. Polyethylene and polypropylene feedstocks have curated the most attention, with demonstrations of pyrolysis^{12–14} and hydroconversion^{15–25} producing light olefins, alkylaromatics, fuel-range hydrocarbons, naphtha, lubricant base oils, and more. Polystyrene (PS) feedstocks have attracted considerably less attention. PS pyrolysis^{26–32} in a N₂ atmosphere has been demonstrated for styrene monomer recovery at high reaction temperatures (400–600 °C). Polycyclic tar oils are a problematic byproduct of pyrolysis due to the poor selectivity of the radical-mediated chemistry.^{33,34} Reactor configurations that incorporate continuous separation of the gas phase (i.e., reactive distillation,

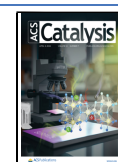
fluidized bed, semicontinuous batch, etc.) can boost styrene yields as high as 70–80%. Kumar et al. demonstrated that the addition of copper and sodium chloride catalysts can further enhance the styrene yield to 84%. Light-activated oxidation of PS^{35–41} to produce oxygenated aromatics (benzoic acid, benzaldehyde, acetophenone, etc.) at ambient conditions has also been demonstrated, but these chemistries must be conducted with solvents and complex catalysts while exhibiting limited selectivity control and low carbon efficiencies. Biosynthetic oxidative routes for PS have also been demonstrated with fungal strains generating pharmacologically relevant molecules,⁴² while engineered bacterium produces oxygenated monomers.⁴³ The hydrogenation of PS to polyvinyl-cyclohexane (PVCH) using nickel, cobalt, and platinum-based catalysts was reported during the mid-20th century.^{44–47} The studies utilized dissolved PS with moderate reaction temperatures (200–300 °C) and high H₂ pressures

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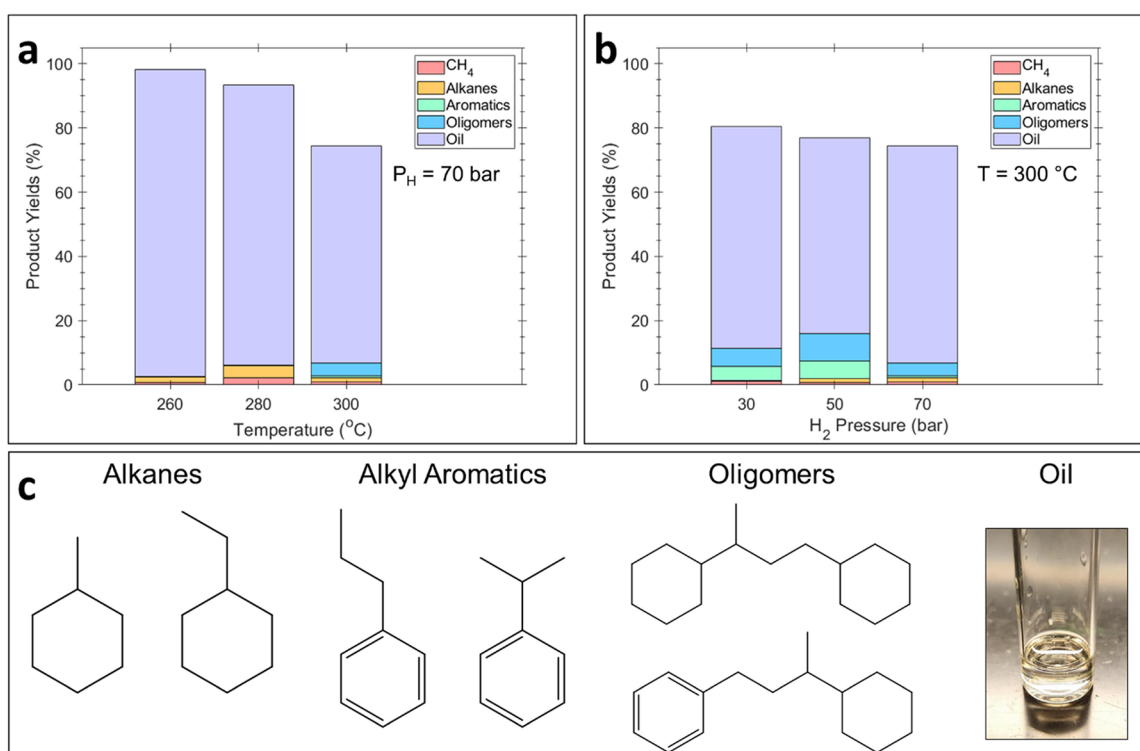


Figure 1. Reaction condition screening for PS97 hydrogenolysis. (a, b) Product yields as a function of reaction temperature and H₂ pressure, respectively. (c) Common alkanes, alkyl aromatics, and oligomers in product fractions, and a picture of product oil. Reaction conditions: 2 g PS97, 100 mg Ni/SiO₂, 2 h, semibatch H₂. Pressures are at the reaction temperature.

(140–265 bar) to achieve complete hydrogenation, often accompanied by polymer molecular weight degradation. Recently, solvent-free PS hydrogenolysis using Ni/SiO₂ was introduced as a proof-of-concept for selectively generating heavy liquid oils at similar temperatures and significantly lower H₂ pressure (30 bar cold).¹⁵ Hydrogenolysis did not produce the byproduct tar oils seen in pyrolysis, highlighting the benefit of hydrogenative catalysts in quenching this undesired pathway. The development of this process is an appealing strategy to create high-value products while expanding the product slate from PS deconstruction.

Herein, we demonstrate the optimized hydrogenolysis of PS with Ni/SiO₂ catalysts for producing high-quality lubricant-base oils. Ni/SiO₂ shows direct reusability for 4 cycles with no deactivation. The chemical structures of these oils and flow properties are analyzed. The reaction network is proposed based on time-dependent experiments with PS and 1,2-diphenylethane as a model molecule. This process is extended to polystyrenes of various molecular weights and commercial PS products. This work expands the scope of polyolefin hydrogenolysis to PS feedstocks and diversifies the existing product slate of PS conversion technologies to include lubricant base oils.

2. RESULTS AND DISCUSSION

2.1. Effect of Reaction Conditions on PS Deconstruction. The reaction temperature and H₂ pressure (P_H) were shown to strongly influence the nickel's catalytic activity for the hydrogenation/hydrogenolysis of PS dissolved in decalin⁴⁶ and low-density polyethylene hydrogenolysis.¹⁵ Solvent-free hydrogenolysis of PS97 ($M_w \approx 97$ kDa, Sigma) was operated in semibatch H₂ to maintain a constant $P_H \pm 2$ bar and washed with methanol to collect light hydrocarbons (i.e., alkanes,

aromatics, and oligomers), and then methylene chloride to collect heavy oils for product collection (see [Methods](#)). [Figure 1a,b](#), [Figure S1](#), and [Table S1](#) display the product yields from the temperature and P_H screening. Catalyst characterization has been reported previously.¹⁵ Excitingly, methane and other gaseous hydrocarbon production is negligible for PS hydrogenolysis, whereas methane is a major byproduct in PE and PP hydrogenolysis.^{15,16,18–20} Minor yields of liquid alkanes, aromatics, and oligomers (<10% each) were detected in the methanol wash. The alkane and aromatic hydrocarbons all contained alkyl chains with 1–4 carbons, while oligomers consisted of fully and partially hydrogenated derivatives of the PS dimer and trimer, per GCMS ([Figure 1c](#)). Heavy oils, which are methanol-insoluble and composed of unreacted and degraded PS, are the dominant products, regardless of reaction conditions. Mass balances span 75–98%, with harsher conditions consistently resulting in less oil and lower mass balances. The unquantified mass is associated with large oligomers (tetramers, pentamers, etc.) that are methanol-soluble but are too heavy to elude during GC analysis (see [Hydrogenolysis Time-Dependence](#)). Unfortunately, little information about the impact of temperature and P_H on the PS97 hydrogenolysis is attainable by purely analyzing the product yields.

Gel-permeation chromatography (GPC) can determine changes in the product oils' molecular weight distributions (MWD). GPC of the neat PS97 shows a trimodal distribution with molecular weights centered at ~ 4.5 kDa, ~ 78 kDa, and ~ 7300 kDa ([Figure 2](#), [Figures S2–S4](#), and [Tables S2–S3](#)) with an overall average molecular weight (M_w) of ~ 97 kDa. Upon PS97 hydrogenolysis at 260 °C and 30 bar H₂, the oil's MWD is effectively the same as the neat PS97 MWD with a slight downward shift in the heaviest mode's center from ~ 7500 to

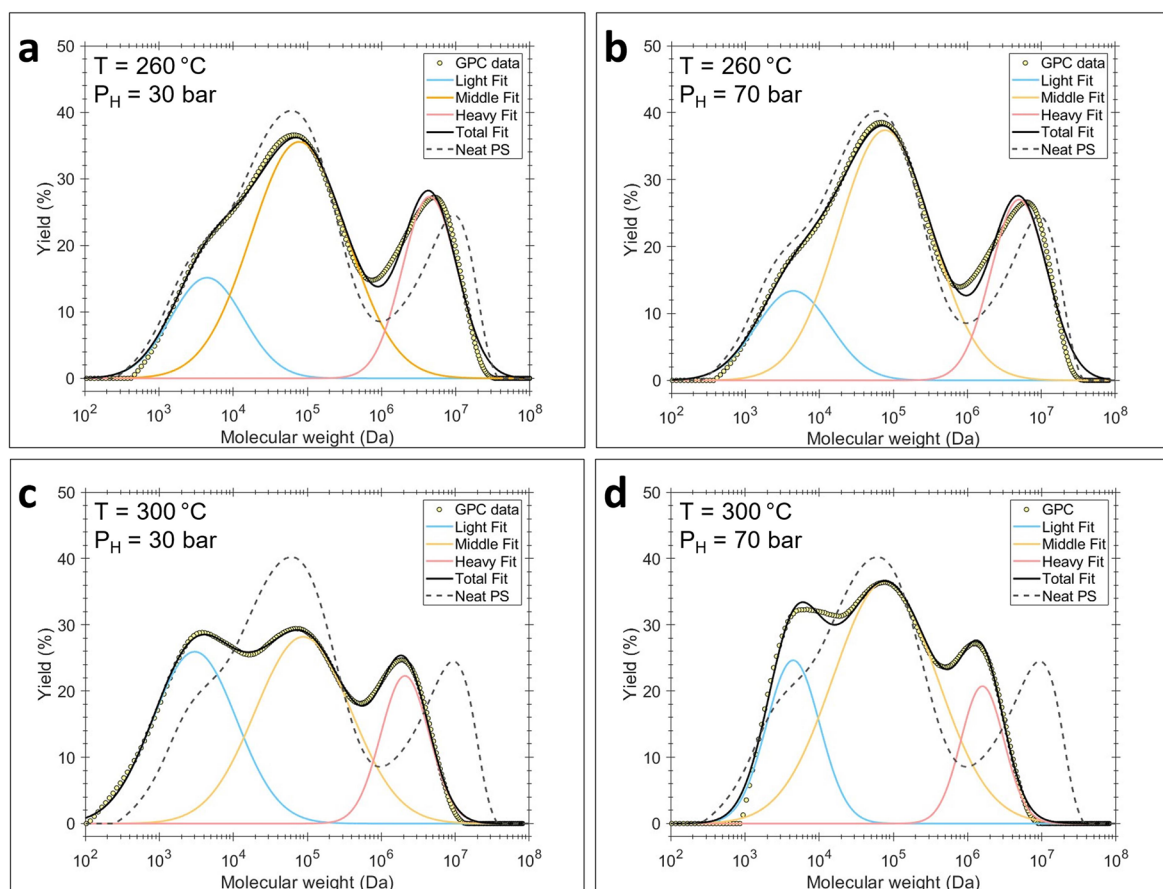


Figure 2. Yield distributions from GPC of methanol-insoluble oils with deconvolutions from the reaction condition screening for PS97 hydrogenolysis. (a) 260 °C and 30 bar H_2 . (b) 260 °C and 70 bar H_2 . (c) 300 °C and 30 bar H_2 . (d) 300 °C and 70 bar H_2 . Reaction conditions: 2 g PS97, 100 mg Ni/SiO₂, 2 h, semibatch H_2 . Pressures are at the reaction temperature.

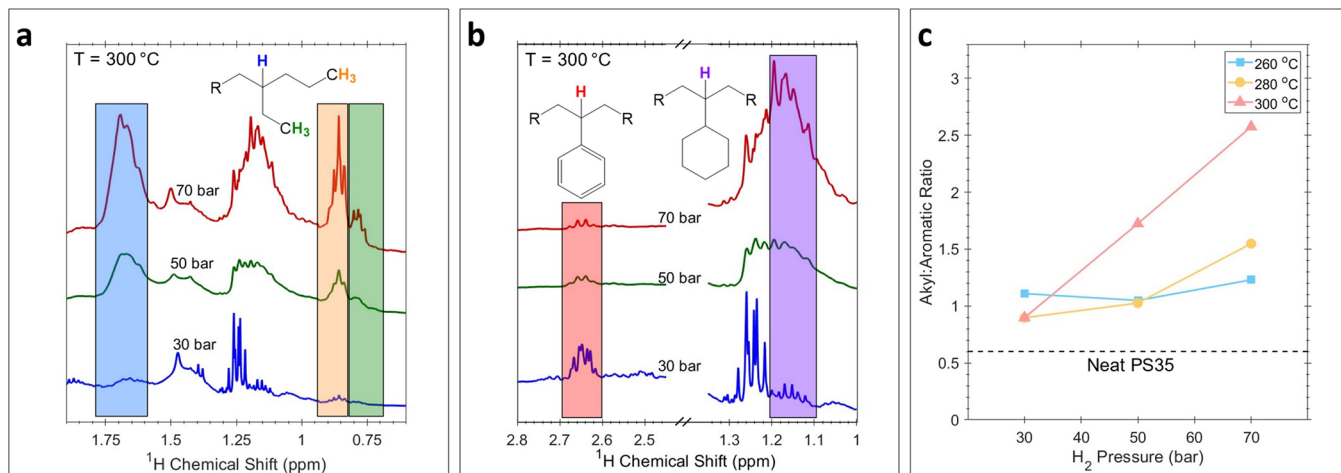


Figure 3. 1H NMR of methanol-insoluble oils from the reaction condition screening for PS97 hydrogenolysis. (a, b) Aliphatic 1H region. (b) Comparison of secondary 1H regions for protons in the polymer backbone adjacent to aromatic and alicyclic chemical environments. (c) AAR for each of the product oils. Reaction conditions: 260–300 °C, 30–70 bar H_2 , 2 g PS97, 100 mg Ni/SiO₂, 2 h, semibatch H_2 . Pressures are at the reaction temperature.

~4600 kDa (Figure 2a, Tables S2 and S3). Increasing P_H to 70 bar while maintaining the reaction temperature shows identical changes in the oil's MWD with a comparable downward shift in the heaviest mode (Figure 2b, Tables S2 and S3). Alternatively, increasing the reaction temperature to 300 °C results in a growth of a ~2.6 kDa mode yield and a concurrent

downward shift in the heaviest mode's position from ~7500 to ~1800 kDa for both pressures (Figure 2c,d, Tables S2 and S3).

Liquid 1H nuclear magnetic resonance (NMR) spectra provide insights into the changes to the PS97 chemical structure upon hydrogenolysis. Product oils from hydrogenolysis at 300 °C show significant differences in the alkyl (0–2.5 ppm) and aromatic regions (6–8 ppm) of the 1H

spectra when increasing P_H from 30 to 70 bar (Figure 3a,b, Figures S5–S7). Figure 3a shows the ^1H protons bound in methine groups from 1.58–1.8 ppm (blue), long-chain primary carbons from 0.82–0.92 ppm (orange), and short-chain primary carbons at 0.66–0.92 ppm (green). Raising P_H shows monotonic growth for these alkyl protons. Methine and primary long-chain groups have the largest relative growth, indicating the hydrogenolysis and hydrogenation of the aromatic ring increases with higher pressures of hydrogen. Additionally, the nonuniformity in the splitting indicates there may be several different components present in the oil product.

Similarly, a broad band spanning 1.0–1.3 ppm also appears in the NMR spectra, becoming the most intense for oils reacted at 70 bar of H_2 (Figure 3a,b). The center of this band spanning 1.1–1.2 ppm is assigned to methine protons adjacent to cyclohexyl-side chains (purple in Figure 3b), based on the reference spectra of PVCH (Figure S8). Examination of the proton adjacent to the aromatic region (2.5–2.8 ppm) corroborates the effect of hydrogen pressure on the concentration of aromatic rings within the polymer (red in Figure 3b). Increasing P_H results in a drop in the abundance of aromatic protons with a concomitant generation of methine protons adjacent to cyclohexyl side chains. These observations provide insight into the PS hydroconversion mechanism. To simplify the analysis while capturing the changes to the PS chemical structure, we define the alkyl-to-aromatic ratio (AAR); the ratio of all aliphatic protons to all aromatic protons (see Methods). The AAR of neat PS97 was 0.6, matching the theoretical value of a polystyrene polymer (Figure 3c). The AAR of the oils from PS97 hydrogenolysis at 300 °C shows a strong increase with P_H , as expected from Figure 3a,b, with a maximum AAR = 2.58 achieved at 70 bar of H_2 . Furthermore, the AAR increases with P_H for the 280 °C oils, but the magnitude is less significant with a maximum AAR = 1.72. Interestingly, P_H does not impact the AAR for the 260 °C oils, with AAR values fluctuating around ~ 1.0 , indicating that there are likely kinetic barriers given that hydrogenation is highly favored by thermodynamics (Table S4).

Overall, this parametric study demonstrates that the product (oils, liquids, and CH_4) yields are unaffected by the temperature and P_H , but the oils are distinct. Both GPC and liquid ^1H NMR elucidate that these parameters strongly influence the molecular weight distribution and chemical structure of the hydrogenolysis oils, respectively. Higher reaction temperatures and P_H facilitate deeper hydrogenation and hydrogenolysis of the PS97; thus, 300 °C and 70 bar of H_2 are primarily used next.

2.2. Thermal Degradation without Ni/SiO₂. Figure 4a and Table S5 show the product yields for reactions at 300 °C and 70 bar of H_2 with and without Ni/SiO₂ to probe the influence of thermal degradation. Reactions without catalyst generate 64% oil yields and 39% yields of methanol-soluble products, comprised mostly of alkylaromatics and oligomers. The oils from hydrogenolysis without catalyst from thermal degradation are black tar-like oils compared to the clear oils attained over Ni/SiO₂. ^1H NMR of the noncatalytic oils reveal a low AAR of 0.5, below the AAR of PS, with large concentrations of aromatic groups still present (Figure 4b and Figure S9). Additionally, the PS conversion was based on ^1H NMR analysis (see Methods) is 2-fold higher for the noncatalytic test than the Ni/SiO₂ (Figure 4b). GPC of the noncatalytic oils unveils a broad molecular weight distribution spanning <0.1 kDa to ~ 2000 kDa with a maximum at ~ 1100

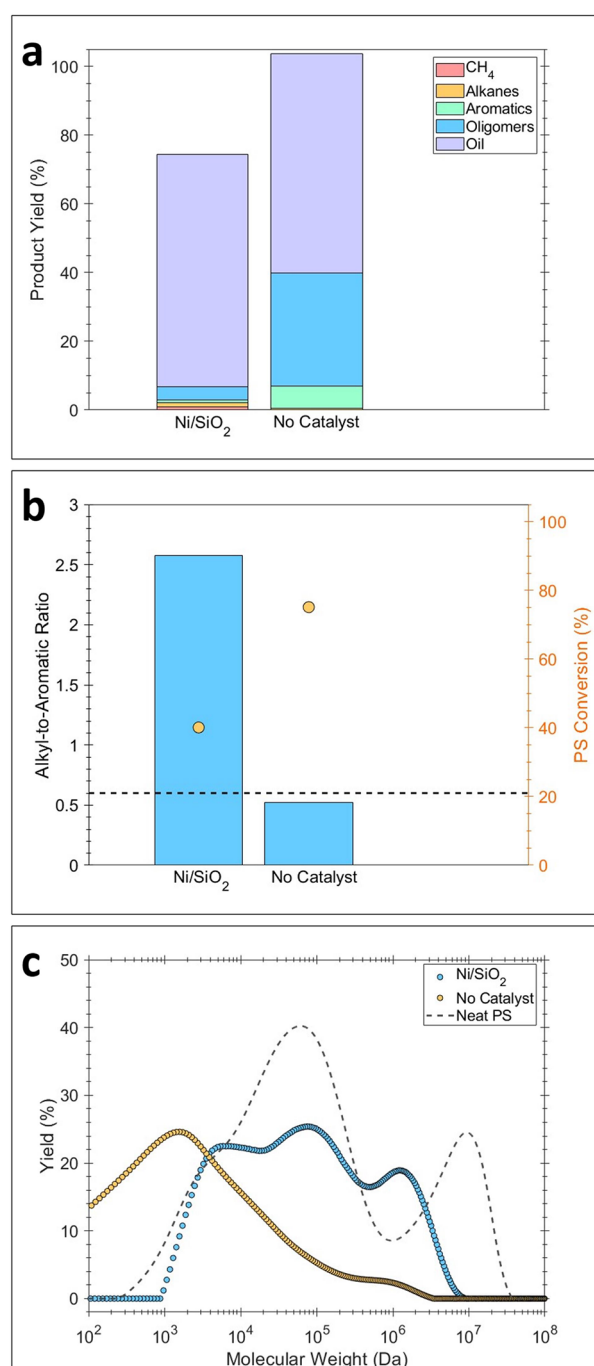


Figure 4. PS97 hydrogenolysis with and without Ni/SiO₂. (a) Product yields. (b) Alkyl-to-aromatic ratio. (c) Yield distributions from GPC of methanol-insoluble oils. Reaction conditions: 300 °C, 70 bar H_2 at reaction temperature, 2 g PS97, 100 mg Ni/SiO₂ or no catalyst, 2 h, semibatch H_2 .

kDa (Figure 4c), exposing massive molecular weight degradation due to thermal reactions. Conversely, the GPC of the catalytic hydrogenolysis oils shows a reduction of the molecular weight of the heaviest molecular weight mode ~ 7500 to ~ 1800 kDa. These results indicate Ni/SiO₂ suppresses thermal degradation reactions of PS97 and shifts the reaction chemistry to purely hydrogenation/hydrogenolysis over the Ni surface. Thermal degradation is much faster than hydrogenolysis for reducing the polymer's molecular weight, but the trade-off is the improved selectivity over Ni/SiO₂.

2.3. Hydrogenolysis Time-Dependence. Time-dependent hydrogenolysis experiments were conducted to explore the P_H effect at 300 °C and gain insight into the reaction network. PS97 hydrogenolysis spanning 0.5–9 h at 30 and 70 bar H_2 are shown in Figure 5 and Table S6. Oil and liquid (i.e., methanol-soluble) product yields are effectively identical between the two P_H values and stable across the reaction span (Figure 5a). However, liquid yields at 70 bar display a substantial increase

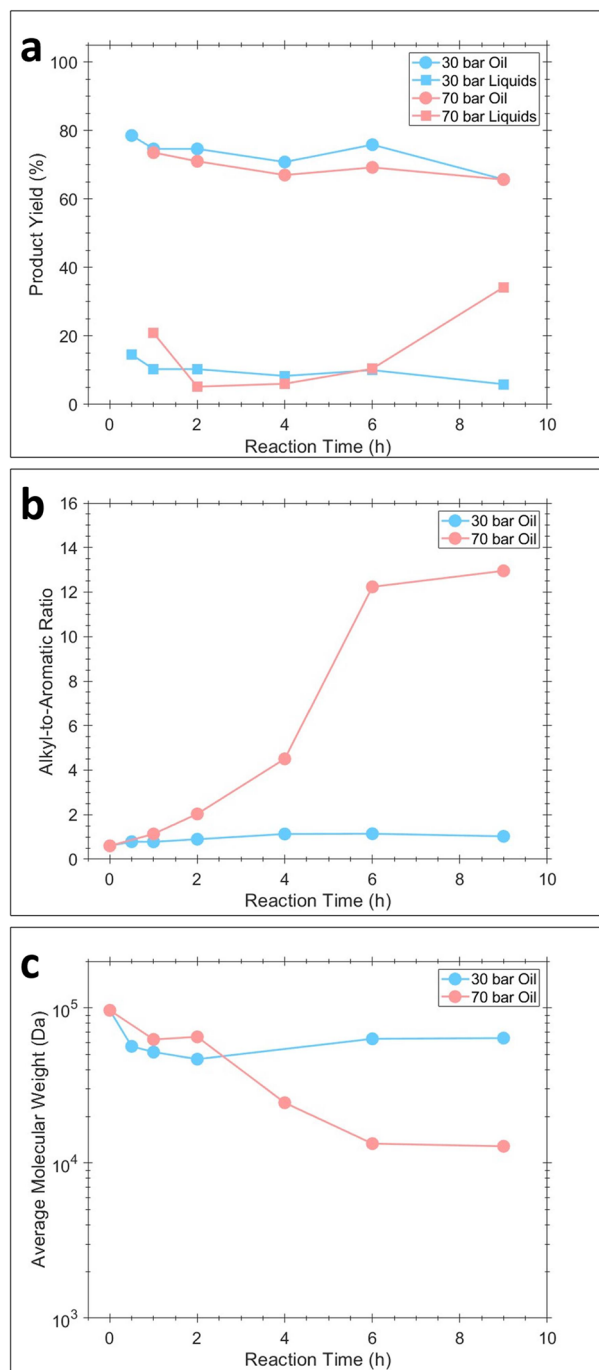


Figure 5. Time-dependent PS97 hydrogenolysis at 30 and 70 bar H_2 . (a) Liquid and oil product yields as a function of reaction time. (b) AAR for the product oils was calculated from 1H NMR. (c) Average molecular weights of the product oils calculated from the GPC. Reaction conditions: 300 °C, 2 g PS97, 100 mg Ni/SiO₂, semibatch H_2 . Pressures are at the reaction temperature.

at 9 h due to the generation of oligomers (Figure S10). This test has full closure of the mass balance, while the average balance for the other time-dependent experiments is ~82%. The quantified oligomer fraction accounts for only the PS dimer, trimer, and their derivatives. The spike in oligomer yields for the 9 h reaction at 70 bar suggests that they form from larger oligomer fragments (tetramers, pentamers, etc.) that are methanol-soluble but too heavy to be eluted during the GC analysis. Akin to the reaction condition screening, product yields provide limited insights requiring oil characterization with NMR and GPC.

Liquid 1H NMR (Figure 5b) and GPC (Figure 5c) reveal a substantial difference in the time-dependent behavior of the AAR and average M_w , respectively, at 30 and 70 bar of H_2 . The AAR of the 30 bar oil has a minor increase from the neat PS value of 0.6 to ~1.1 in 0.5–4 h, with longer reaction times showing little change (Figure 5b and Figure S11). The MWD of the oil closely resembles that of the neat PS97 with a downward shift in the heaviest mode's center from ~7300 to ~1600 kDa over the entire reaction span (Figure S12). Deconvolution of the MWD indicates minor redistribution of the weight fractions for the two lightest modes (~4.5 and ~77 kDa), while the heaviest mode's weight fraction is stable (Figure 6a). Overall, the MWD changes culminate to a moderate reduction in the average M_w from ~97 to ~57 kDa for the 30 bar oil, that is achieved within 0.5 h before plateauing at longer times (Figure 5c). The low, stationary AAR coupled with the stagnant average M_w expose that 30 bar H_2 is insufficient for Ni/SiO₂ to facilitate hydrogenation and hydrogenolysis.

Conversely, the 70 bar oils display a linear growth in the AAR to 4.5 in 1–4 h and 12.2 at 6 h (Figure 5b and Figure S13). The MWD also demonstrates significant changes in the 1–6 h time span as the weight fraction of the heaviest mode (~2300 kDa) is gradually reduced from ~20 to 0 wt %, with a concurrent growth of the lightest mode (~4.5 kDa) from ~25 to ~60 wt % (Figure 6b and Figure S14). The elimination of the heaviest mode results in a substantial reduction in the average M_w to ~13 kDa in 6 h (Figure 5c). Extending the reaction time to 9 h shows minimal improvement in the AAR (Figure 5b), MWD weight fractions (Figure 6b), and average M_w (Figure 5c). The resulting oils at 6 and 9 h are completely bimodal and have effectively identical MWD (Figure 6c and Figure S14).

These findings demonstrate the importance of high P_H for Ni/SiO₂ to facilitate PS97 hydrogenolysis. Hydrogenolysis of the PS backbone occurs only after significant hydrogenation of the aromatic rings, suggesting a sequential reaction network. Furthermore, the reduction in the average M_w is dominated by the deconstruction of the heaviest polymer chains ($M_w > \sim 100$ kDa), while lighter molecular weight chains show limited deconstruction. The lower reactivity of these lighter chain fragments may stem from weaker surface adsorption that results in short surface residence times. Reaction rate analysis (see Probe Reactions with 1,2-Diphenylethane (DPE)) reveals that C–C bond scission rates are 1000× slower than aromatic ring hydrogenation. These slow reaction rates demand long surface residence times, which are unlikely for heavier molecular weight species. The presence of strong surface adsorption has been revealed for polyolefin systems,^{15,23,48,49} with major mechanistic insights, such as the adhesive isomerization cycle²³ and the divergent hydrogenolysis

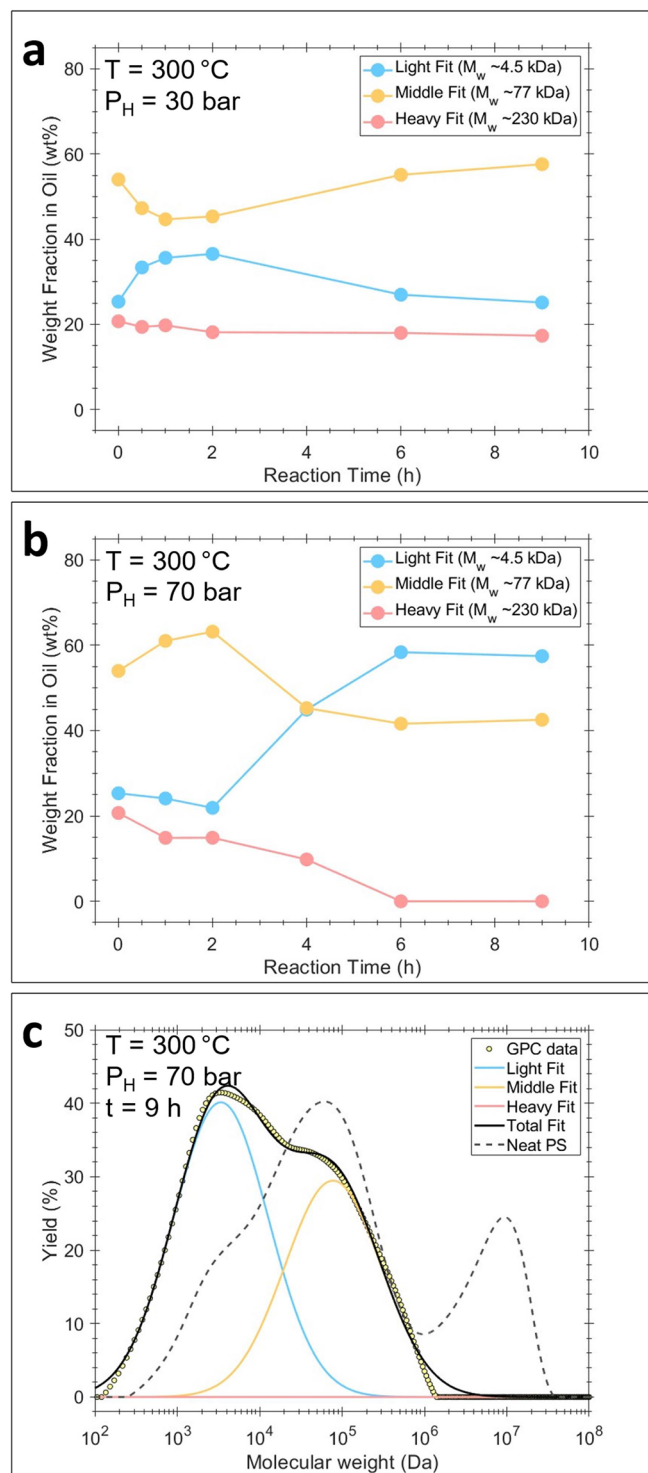


Figure 6. GPC analysis for the oils from the time-dependent PS97 hydrogenolysis at 30 and 70 bar H_2 . (a, b) Weight fractions for the three Gaussian modes fit to the GPC yield curves at 30 and 70 bar H_2 , respectively. (c) Yield distributions of the product oils after 9 h of reaction time at 30 and 70 bar of H_2 , respectively. Reaction conditions: $300\text{ }^{\circ}\text{C}$, 2 g PS97, 100 mg Ni/SiO₂, semibatch H_2 . Pressures are at the reaction temperature.

mechanism¹⁵ in PE hydrocracking and hydrogenolysis, respectively.

The hydrogenolysis oils from the long reaction times (6–9 h) at $300\text{ }^{\circ}\text{C}$ and 70 bar of H_2 were clear and flowed easily, suggesting these oils are good candidates for lubricant base-oil

applications. Table 1 shows the measured properties of the PS97 oils from the 6 h reaction compared to a commercial synthetic lubricant base oil and oils obtained from polypropylene (PP) hydrogenolysis.²² High-quality lubricants typically demand low pour points, high oxidation onset temperature, and high viscosity indices (VI) as these reflect the oil's ability to flow at low temperatures, its thermal stability, and the viscosity dependence on temperature, respectively. The VI is a primary parameter for classifying lubricant qualities, with lower-grade Group I–III oils having $\text{VI} < 120$ and high-grade Group IV oils having $\text{VI} > 120$.^{50,51} The PS97 oil has a high VI of 130 ± 2 , qualifying it as a Group IV oil. This VI is lower than but comparable to those of the reported PP oils and the Chevron PAO6. The pour point and oxidation onset for the PS97 oil outperform those of the PP oils but are inferior to those of commercial PAO6. The kinematic viscosity (KV) of PS97 oil is approximately double that of the light PP oil and Chevron PAO6, meaning that these oils are not suitable for identical applications. High KV lubricants are commonly used in heavy machinery applications. However, flow additives could be used to tailor KV and VI of the PS97 oils to be more comparable to PAO6. Overall, these properties identify lubricant base oils as attractive applications for PS97 oils.

2.4. Probe Reactions with 1,2-Diphenylethane (DPE).

DPE was used as a model molecule to probe the PS97 hydrogenolysis reaction mechanism. Three-phase continuous flow experiments reveal that DPE consumption rates have Arrhenius behavior with a strong P_{H} dependence, like PS97 hydrogenolysis (Figures S15 and S16). Unfortunately, full DPE conversion in continuous flow was difficult to attain due to the high H_2 stoichiometry (6 mol of H_2 to 1 mol of DPE) that resulted in rapid H_2 depletion. Time-dependent experiments in the semibatch H_2 system were first run instead to map the reaction progression (Figure 7a) at $300\text{ }^{\circ}\text{C}$ and 50 bar H_2 . Initial semibatch H_2 experiments revealed that DPE was substantially more reactive than PS97, requiring the Ni/SiO₂ loading to be limited to 5–20 mg and increasing the substrate mass to 3 g to access the kinetic envelope. Time-dependent experiments reveal that the aromatic rings of DPE are sequentially hydrogenated to produce 1,2-cyclohexylbenzyl-ethane (CHBE) and then 1,2-dicyclohexylethane (DCHE) (Figure 7a,b, Figure S17). Surprisingly, only minor quantities of methylcyclohexane (MCH) were detected and toluene (TOL) or products from ring-opening reactions were not detected. Longer DPE residence times (τ) increase the MCH production. Comparison of initial rates demonstrates that DPE and CHBE hydrogenation rates are approximately identical ($r \approx 10\text{ mmol}_{\text{arene}}\cdot\text{min}^{-1}$), whereas the MCH production (i.e., DCHE hydrogenolysis, $r \approx 0.01\text{ mmol}_{\text{C-C}}\cdot\text{min}^{-1}$) is 1000× times slower. These rate differences could justify the MWD stability of the PS97 oils after long times but conflict with the relatively fast degradation of the heaviest PS modes after hydrogenation (Figures 5 and 6). Elias and Etter⁴⁶ reported on the hydrogenation of dissolved PS to PVCH with a final $M_w \sim 62\text{ kDa}$ using the Raney-Ni catalyst and proposed that the PS molecular weight degradation was due to thermal and mechanical effects rather than direct hydrogenolysis. The DPE rate analysis and PS97 hydrogenolysis reactions support these findings; yet, it is unclear why chains below $M_w \approx 100\text{ kDa}$ are less prone to deconstruction.

Interestingly, Ni/SiO₂ has previously been shown to facilitate internal C–C bond cleavage of *n*-hexacosane (*n*C26) with hydrogenolysis rates of $\sim 0.1\text{ mmol}_{\text{C-C}}\cdot\text{min}^{-1}$.¹⁵

Table 1. Analysis of Oil Properties From PS97 Hydrogenolysis and Comparison with Commercial Lubricant Base Oils and Previously Reported Oils from Plastic Waste Hydrogenolysis

sample	pour point (°C) ^a	oxidation onset (°C) ^b	KV40 (cSt) ^c	KV100 (cSt) ^c	viscosity index ^d
PS97 Oil (6 h)	−30	184	79.8 ± 0.9	11.2 ± 0.2	130 ± 2
Chevron PAO6 ^e	−62	246	30.8	5.9	139
light PP Oil ^f	−15	179	30.3 ± 0.2	6.2 ± 0.1	146 ± 7
heavy PP Oil ^f	−18	175	113.5 ± 0.8	42.2 ± 0.3	149 ± 7

^aASTM D5949. ^bASTM E2009. ^cKV40 and KV100 are kinematic viscosities measured at 40 and 100 °C, respectively. ^dCalculated from KV40 and KV100. ^eCommercial synthetic base oil from Chevron; oil properties are from the manufacturer's specifications sheet. ^fOil properties reported from PP hydrogenolysis with Ru/TiO₂ in ref.²² Reaction conditions: 300 °C, 70 bar H₂, 2 g PS97, 100 mg Ni/SiO₂, 6 h, semi-batch H₂.

There is an order of magnitude difference in the internal C–C bond hydrogenolysis rates between nC26 and DCHE (i.e., the rate of DCHE consumption), suggesting that the carbon structure of DCHE is unable to form the same surface ensembles of nC26, forcing DCHE hydrogenolysis through a higher energy transition state.⁵² Moreover, nickel's sensitivity to the hydrocarbon structure and limited ring-opening capacity is well-established.^{53–56} This surface ensemble effect also rationalizes why ring-opening reactions are not observed in DPE hydrogenolysis, and the cyclohexyl side chains in the PS97 oils remain intact.

2.5. Ni/SiO₂ Catalyst Reuse. Catalyst reuse is a valuable metric in determining the industrial viability of a catalyst.⁵⁷ Spent Ni/SiO₂ was collected after the 6 h reactions at 300 °C and 70 bar of H₂ by filtration and dried in air at 25 °C before testing for direct reusability for PS97 hydrogenolysis. Figure 8a shows that the oil and liquid product yields are stable over four reuse cycles. Conversely, the oils' AAR from liquid ¹H NMR increases from 12.2 to 18.9 over the first two reuses and stabilizes at further reuses. The improved AAR over each reuse demonstrates that Ni/SiO₂ does not deactivate; on the contrary, its hydrogenation performance improves with each cycle. Thermogravimetric analysis of the Reuse-4 catalysts indicates minimal coking (<1 wt %) on the catalyst surface (Figure S18). Comparing the X-ray diffraction (XRD) of fresh and Reuse-4 Ni/SiO₂ reveals partial restructuring of the Ni nanoparticle morphology from the Ni(200) facet at 44.6° to the Ni(111) facet at 52.0° (Figure 8b). The Ni(111) facet is the primary active phase in benzene hydrogenation as it readily activates the C–H bond due to the close symmetry in the Ni lattice spacing and C–C aromatic bond length.^{58,59} Subsequently, the increased Ni(111) fraction rationalizes the improved activity upon reuse.

2.6. Feedstock Variation. Other virgin and commercial PS feedstocks were used to probe the versatility of Ni/SiO₂ for PS hydrogenolysis at 300 °C, 70 bar of H₂, and 9 h. Hydrogenolysis of virgin PS5300 and PS6000 feedstocks produces 67 and 69% oil yields, respectively, comparable to 66% of PS97 (Figure 9a). Interestingly, liquid product yields are >25% lower for PS5300 and PS6000 due to a reduction in oligomers likely due to the heavier *M_w* of these feedstocks that produce heavy oligomers (tetramers, pentamers, etc.) not detectable by GC. The high AAR of the PS5300 and PS6000 oils indicates considerable hydrogenation (Figure 9b and Figure S19). However, there is not a clear trend between AAR and molecular weight as the PS5300 oil's AAR is comparable to that of the PS35 oils while the PS6000 oil's AAR is ~50% lower than the latter two, demonstrating a lower activity for PS6000. The differences in the virgin PS reactivities could stem from molecular weight, architecture, or viscosity,^{21,60} and slip

agents or other performance additives may be present and inhibit the deconstruction.^{17,25}

Hydrogenolysis of commercial PS products (Figure 9c) shows a wide variability in reactivity. Rigid PS products (i.e., Petri dish and F-tube) produce clear, inviscid oils with high AARs, indicating significant hydrogenation and deconstruction (Figure 9d). Conversely, the flexible W-boat produces black tar-like oils, while the foam product generates an auburn, brittle solid (Figure 9d) and considerable yields of alkylaromatics (9%) and oligomers (33%). The low AAR for the W-boat oils and foam solids after reaction reveals limited hydrogenation, with the foam solid's AAR matching that of neat PS (Figure 9b and Figure S19). The deactivation of Ni/SiO₂ is associated with the performance additives in these PS products (Figure 4). X-ray fluorescence (XRF) of the W-boat identified a significant quantity (~5 wt %) of calcium likely as fine calcium carbonate (CaCO₃) nano/microparticles that are commonly added to plastics as a filler and mechanical property modifiers (i.e., improve PS flexibility).⁶¹ The surface of CaCO₃ particles is also often functionalized with organic and inorganic components to improve dispersion within the polymer via colloidal interactions and further tune mechanical properties.⁶¹ Vacuum filtration, centrifugation, decanting, and other separation methods could not isolate the colloiddally suspended CaCO₃ particles from the PS polymer. We suspect the CaCO₃ particles or the surface's functionalized components are responsible for the Ni/SiO₂ deactivation. Alternatively, expanded/extruded foams contain blowing agents, dispersion agents, foam stabilizers, and other additives to generate the porous foam structure.⁶² These components could be responsible for the Ni deactivation as they can contain halogen, hydrazine, sulfonyl, and other organic functionalities.⁶²

3. CONCLUSIONS

In this study, we demonstrate nickel supported on silica (Ni/SiO₂) for polystyrene (PS) hydrogenolysis to lubricant base oils under mild conditions (300 °C and 70 bar of H₂) within 6 h. Unlike polyolefins, the yield to methane and other gaseous hydrocarbons is small. Process optimization revealed that the yields of gas, liquid (i.e., alkanes, aromatics, and dimer and trimer oligomers), and oils are unaffected by temperature and H₂ pressures (P_H). However, GPC and NMR analysis of the oils reveal a reduction in the molecular weight distribution and significant hydrogenation of the aromatic rings in PS, respectively, at higher temperatures and P_H. Furthermore, we defined the alkyl-to-aromatic ratio (AAR) from ¹H NMR characterization that consolidates the complex structural changes of PS into a single parameter to simplify the oil analysis. Time-dependent PS hydrogenolysis experiments at 30 and 70 bar H₂ further demonstrate the strong dependence on

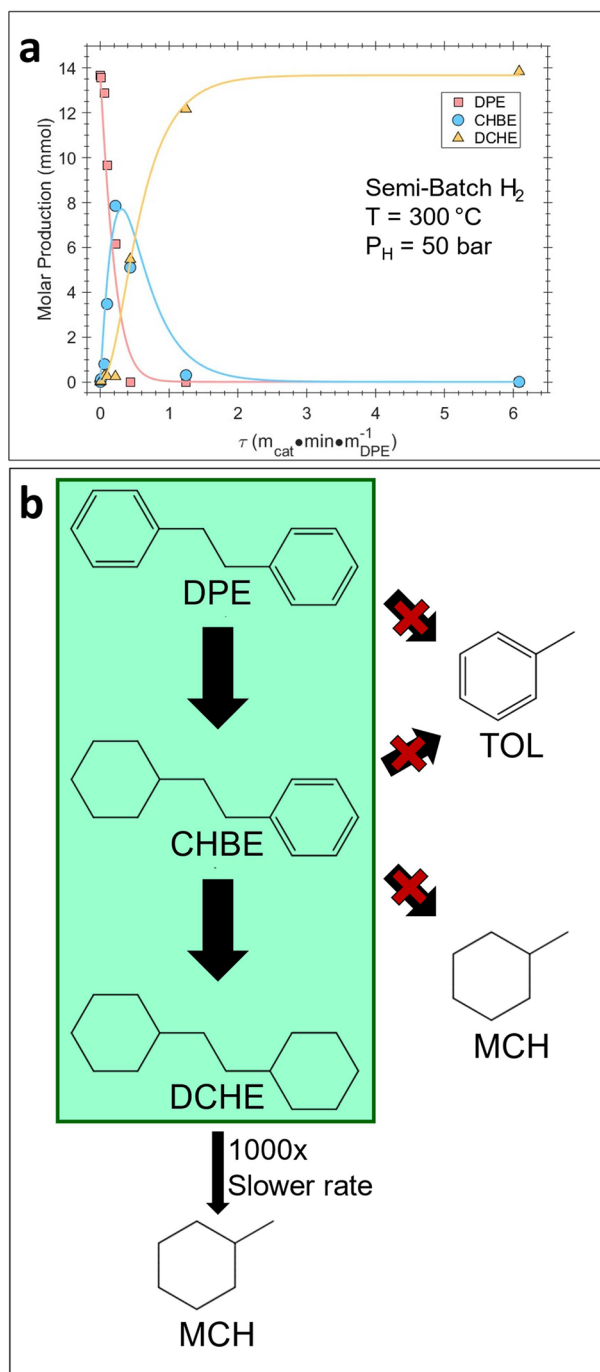


Figure 7. Hydrogenolysis kinetics of probe molecule 1,2-diphenylethane (DPE). (a, b) Kinetic evolution of major reaction products in a semibatch H_2 reactor and respective DPE hydrogenolysis reaction pathway. Reaction conditions: 300 $^{\circ}C$, 50 bar H_2 at reaction temperature, 3 g DPE, 5–20 mg Ni/SiO₂, 0.5–6 h, semibatch H_2 .

P_H , while revealing that hydrogenolysis occurs sequentially, requiring significant aromatic ring hydrogenation prior to reduction of the molecular weight. Rate analysis for 1,2-diphenylethane (DPE) hydrogenolysis exposes that aromatic ring hydrogenation is 1000x faster than C–C bond cleavage over Ni/SiO₂, rationalizing the PS hydrogenolysis reaction pathway. After 6 h reactions, the oils show Group IV lubricant properties comparable to commercial synthetic oils and better to base oils produced from polypropylene hydrogenolysis. Lastly, Ni/SiO₂ catalysts are amenable to managing other

virgin feedstocks and rigid commercial PS products, whereas flexible PS and packing foam products cause deactivation associated with the presence of additives. These results demonstrate high-quality lubricant base oils from PS using Ni/SiO₂, expanding the product slate for PS deconstruction technologies while exposing the importance of earth-abundant metals in the transition toward a circular economy.

4. METHODS

4.1. Catalyst Synthesis. Nickel nitrate hexahydrate (Aldrich, Ni(NO₃)₂•6H₂O, $\geq 99.999\%$ trace metal basis) was used as received and silica-gel (Sigma-Aldrich, SiO₂, Davisil grade 646, 35–60 mesh) was calcined in static air at 650 $^{\circ}C$ for 3 h (2 $^{\circ}C/min$ ramp) prior to synthesis. 15 wt % Ni/SiO₂ was prepared via wetness impregnation of a nickel nitrate hexahydrate solution, dried in air at 110 $^{\circ}C$, and then calcined at 550 $^{\circ}C$ for 3 h (2 $^{\circ}C/min$ ramp). Ni/SiO₂ was crushed and sieved to 63–125 μm (120–230 mesh) for reactions with 1,2-diphenylethane (DPE).

4.2. Catalyst Characterization. Ni/SiO₂ was characterized previously¹⁵ using X-ray diffraction, N₂ sorption, transmission electron microscopy, and N₂O titrations.

4.3. Polystyrene Hydrogenolysis Reactions and Analysis. All polystyrene samples from Aldrich were used directly without any preprocessing. Commercial polystyrene products were cut into ~ 1 cm² pieces prior to the reaction. Information about the polymer feedstocks is summarized in Table 2. Other chemicals and consumables used for product extraction and characterization are n-dodecane (C₁₂H₂₆, Thermo Scientific, 99%), methanol (CH₃OH, Fisher Chemical, HPLC grade, $\geq 99.9\%$), methylene chloride (CH₂Cl₂, Fisher Chemical, HPLC grade, $\geq 99.9\%$), tetrahydrofuran (C₄H₈O, Fisher Chemical, $\geq 99.9\%$, unstabilized), chloroform-d (CDCl₃, Thermo Scientific, > 99.8 atom % D, 0.02–0.03 v/v % TMS), and Whatman filter paper (GE, 11 μm).

Hydrogenolysis reactions were conducted in a 50 mL Parr stainless steel batch reactor with a pressure gauge, thermocouple, and 25.4 mm $\times$ 8 mm PTFE magnetic stir bar. A band heater connected to a PID controller was used to heat the reactor and maintain the reactor temperature. The reactor and band heater were wrapped in fiberglass insulation to decrease the ramping time and improve temperature stability.

Prior to reactions, the Ni/SiO₂ catalyst was reduced in 100 mL/min equimolar He/H₂ at 450 $^{\circ}C$ for 2 h (10 $^{\circ}C/min$ ramp). All catalytic reactions were conducted solvent-free. A total of 100 mg of catalyst was added with 2.0 g of PS substrate to the Parr reactor and the contents were well-mixed using a stir bar. The Parr reactor was sealed, connected to a calibrated semibatch H₂ line (Scheme 1), and purged six times with 15 bar H₂. The reactor was pressurized to a cold reactor pressure (18–40 bar) and then heated to the reaction temperature (260–300 $^{\circ}C$). Stirring was turned on at 500 rpm when the reactor reached 240 $^{\circ}C$. Reactions were run for specified intervals (0.5–9 h) after the reactor reached reaction temperature. Additional H₂ was delivered to the reactor via a semibatch line when the reactor fell ~ 2 bar below the desired reaction pressure. The needle valve to the H₂ cylinder and the ball valve would be opened to charge the dosing line to the regulator pressure. The needle valve to the H₂ cylinder was then closed. The initial pressure of the dosing line was recorded, and then H₂ was dosed to the Parr reactor by opening the reactor's needle valve until the reactor's pressure

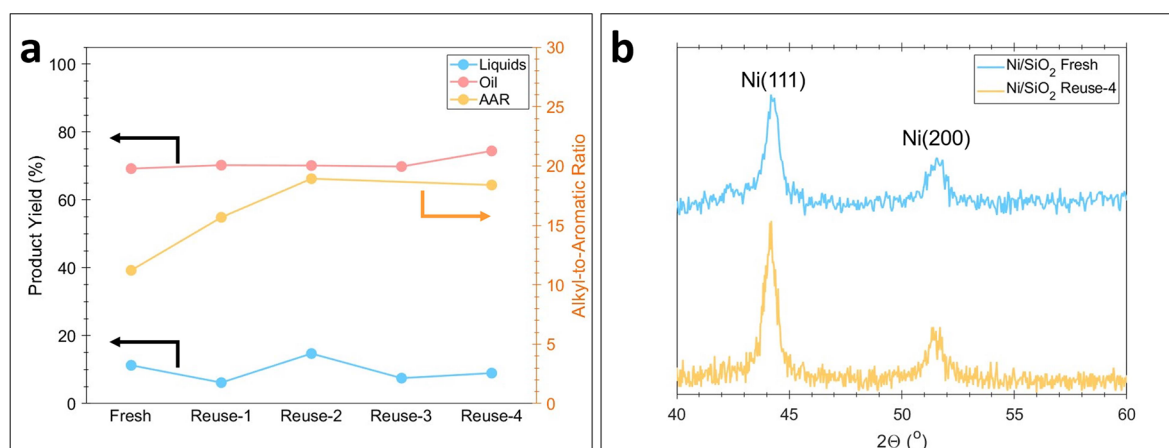


Figure 8. Direct Ni/SiO₂ reuse for PS97 hydrogenolysis. (a) Liquid and oil product yields with the oil AAR calculated from ¹H NMR. (b) XRD catalyst of the fresh and reuse-4 Ni/SiO₂ catalyst. Reaction conditions: 300 °C, 70 bar H₂ at reaction temperature, 2 g PS97, 100 mg catalyst, 6 h, semibatch H₂.

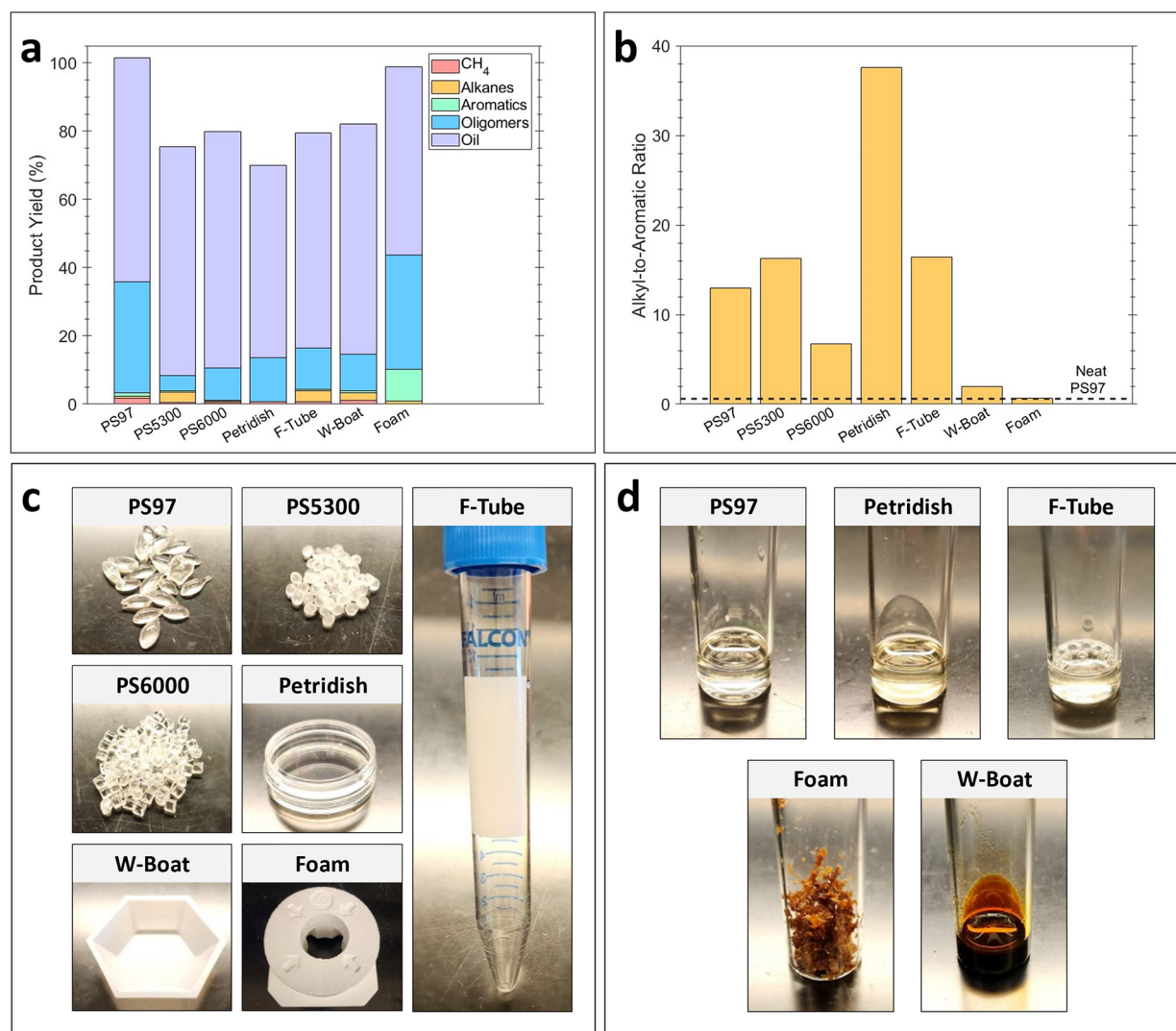


Figure 9. Hydrogenolysis feedstock variation including virgin and commercial polymer products. (a) Product yields. (b) AAR for the product oils calculated from ¹H NMR. (c, d) Pictures of raw PS feedstocks and resulting products from hydrogenolysis reactions, respectively. Reaction conditions: 300 °C, 70 bar H₂ at reaction temperature, 2 g substrate, 100 mg Ni/SiO₂, 9 h, semibatch H₂.

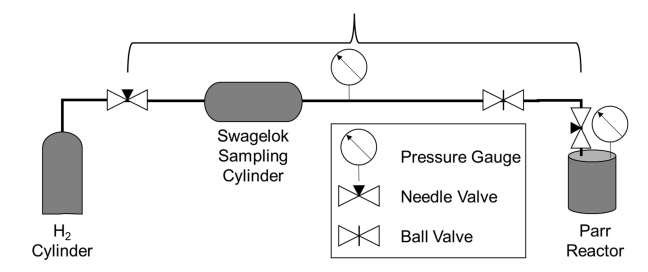
reached the desired reaction pressure. The final pressure of the dosing line was recorded and the quantity of dosed H₂ was

back-calculated (Figure S20, eqs S1–S4) using the pressure drop in the dosing line. Upon reaching the reaction time was

Table 2. Polymer Feedstocks Used in Hydrogenolysis Reactions

feedstock classification	polymer	vendor	molecular weight from vendor (kDa)	molecular weight from GPC analysis ^a (kDa)	referencing in text
virgin granules	polystyrene	Sigma	~35	~97	PS97
	polystyrene	Sigma	~180	~5300	PS5300
	polystyrene	Sigma	~280	~6000	PS6000
commercial polystyrene products	weigh boat	Fisherbrand			W-boat
	Petri dish	Fisherbrand			Petri dish
	Falcon tube	Corning Falcon			F-tube
	expanded polystyrene foam	ULINE			foam

^aGel-permeation chromatography of commercial PS products was not conducted due to additives that could potentially damage the instrument.

Scheme 1. Semi-Batch H₂ Dosing System Schematic

reached, the reactor was quickly quenched in an ice bath. Product collection was conducted when the reactor reached 2 °C. Gas samples were collected from the Parr reactor headspace by charging into a 1 L Tedlar gas sampling bag and analyzed with GC-FID (Agilent HP-Plot/Q+PT GC column). The reactor was opened and washed with ~20 mL of methanol containing 20 mg of n-dodecane as an internal standard to collect light hydrocarbons and oligomer fragments. This methanol mixture was separated from the solids via filtration, and products were quantified using GC-FID (Supelco Vocol and Agilent HP-5 columns) and identified using GC-MS (Agilent DB-1 column). Calibration coefficients and retention times for light products were measured by injection of hydrocarbons (Hydrocarbon Standard, 98537, Absolute Standards) and aromatics multimix standards (Volatiles Mix #1, 94855, Absolute Standards) (Figure S21, Tables S7 and S8). Figure S22 contains a representative chromatogram of methanol-soluble reaction products.

The remaining oils and unreacted PS were collected with ~20 mL of methylene chloride and filtered. Methylene chloride was removed by using a rotary evaporator, and the oil mass was measured gravimetrically. A mass balance (MB) was conducted as follows:

$$MB = \frac{m_g + m_L + m_{oil}}{m_{PS,i}} \quad (1)$$

where m_g , m_L , m_{oil} , and $m_{PS,i}$ is the mass of gas, light hydrocarbons soluble in methanol, oils, and unreacted PS dissolved in methylene chloride, and the initial mass of polystyrene, respectively. Product yields for group i was calculated as

$$Y_i = \frac{m_i}{m_{PS,i}} \times 100\% \quad (2)$$

Oil products were analyzed using room temperature gel permeation chromatography (GPC) using Styragel HR 4, HR 3, and HR 0.5 columns (dimensions of 4.6 × 300 mm) in

tandem using THF as solvent (150 μL/min) and a Waters 2414 refractive index detector (RID). The retention times were calibrated using a Polystyrene Standards Kit (Waters, WAT058931, 0.500 kDa ≤ M_w 2500 Da). Yield distributions (y_o) were calculated as

$$y_o = Y_o H_N(t) = Y_o \frac{H(t)}{\int H(t) dt} \quad (3)$$

where Y_o and $H_N(t)$ are the oil yield and normalized RID response, respectively. Deconvolution of yield distributions as a function of the logarithm of molecular weight were conducted using a custom MATLAB program. Briefly, the number of desired deconvoluted peaks and initial guesses for the peak's center (x_o), maximum intensity (I_{max}), and full-width at half-maximum (FWHM) were input. An iterative least-squares method was employed to minimize the residuals between the fit's envelope and the yield distribution by adjustment of the three peak parameters. Additionally, the program could hold any or all of the peak parameters for a given peak when solving the peak deconvolution problem. Deconvoluted peaks were assumed to have Gaussian peak shapes, described by

$$f_i(\log(M_w)) = I_{max} \exp \left[-\frac{(\log(M_w) - x_o)^2}{2 \left(\frac{FWHM}{2\ln(2)} \right)^2} \right] \quad (4)$$

GPC of the light oil fractions would occasionally become convoluted with the elution of the THF mobile phase. Deconvolution of the oil's yield distribution was adapted by adding a fourth peak to model the elution of the THF, while the lightest mode's peak center was fed an initial guess of the average peak position for this mode from other fittings ($x_o = 3.65$), and the range of values was constrained.

The chemical structure of the oil was analyzed using liquid state NMR operating at a proton frequency of 400 MHz. NMR samples were prepared by adding ~40 mg of oil dissolved in 800 μL of chloroform-*d*. A standard ¹H quantitative NMR sequence was used with averaging over 16 scans. Integrations were done using the TopSpin software to calculate the alkyl-to-aromatic ratio (AAR) defined as

$$AAR = \frac{A_{Alkyl}}{A_{Aro}} \quad (5)$$

where A_{Alkyl} is the integrated region from 0 to 2.5 ppm and A_{Aro} is the integrated region from 6 to 8 ppm. The contribution of chloroform-*d* at 7.24 ppm was not included in the integration

for the AAR calculation. A subsequent ^{13}C NMR experiment was run using a standard composite-pulse decoupling pulse sequence averaged over 1024 scans. This was used to calculate the PS conversion (X_{PS}), which is based on the presence of the methine carbon adjacent to an aromatic group in the PS carbon backbone which appears at 41 ppm.⁶³ Specifically, conversion is calculated as

$$X_{\text{PS}} = \frac{A_{41\text{ppm}}^i - A_{41\text{ppm}}^f}{A_{41\text{ppm}}^i} \times 100\% \quad (6)$$

where $A_{41\text{ppm}}^i$ is the normalized area of the methine carbon in the PS97 sample and $A_{41\text{ppm}}^f$ is the normalized area of the methine carbon in the oil sample. All areas in eq 6 are normalized to the sample's mass used in the NMR analysis.

The KV40 and KV100 of the product oils were measured using a Cannon 200 microviscometer. The viscometer was heated by using a stirred mineral oil bath at the desired temperature (40 or 100 °C). For measurements, 200 μL of sample were added to the viscometer and vacuum was applied on the capillary side to raise the oil above the measurement zone. Care was taken to ensure that bubbles were not present in the viscometer capillary. The vacuum was then removed to allow the oil to fall, and the time span for the sample to flow through the measurement zone was recorded in triplicate. Cannon N44 was used as the viscometer standard.

Pour points and oxidation onset temperatures were measured by Southwest Research Institute according to ASTM D5949 and ASTM E2009 standardized protocol, respectively.

4.4. 1,2-Diphenylethane (DPE) Hydrogenolysis Reactions and Analysis. 1–2-Diphenylethane ($\text{C}_{14}\text{H}_{14}$, TCI America, $\geq 99.0\%$) was used as received. Other chemicals and consumables used for product extraction and analysis are *n*-dodecane ($\text{C}_{12}\text{H}_{26}$, Thermo Scientific, 99%), *n*-hexacosane ($\text{C}_{26}\text{H}_{54}$, Sigma, $\geq 99\%$), methylene chloride (CH_2Cl_2 , Fisher Chemical, HPLC grade, $\geq 99.9\%$), and Whatman filter paper (GE, 11 μm).

Ni/SiO_2 was crushed and sieved to 63–125 μm fraction for all reactions with 1,2-diphenylethane (DPE). Batch hydrogenolysis reactions followed the same procedure as the polystyrene experiments, but 5–100 mg of sieved Ni/SiO_2 were added to the Parr reactor with 3 g DPE and a PTFE stir bar. The reactor was connected to the semibatch H_2 line, purged six times at 15 bar H_2 , charged to 38 bar H_2 , and began heating to 300 °C. Stirring was set to 500 rpm when the reactor hit 75 °C. Reactions were maintained for specified intervals (1–6 h) after reaching reaction temperature, with additional H_2 being delivered via the semibatch line when the reactor pressure reached 48 bar. Reactions were stopped by quenching in an ice bath.

Gas samples were collected using a Tedlar gas bag when the reactor temperature reached 2 °C and analyzed with GC-FID (Agilent HP-Plot/Q+PT GC column). Liquid/solid reaction products were collected with ~ 20 mL of methylene chloride and 20 mg of dodecane. This solution was diluted 50-fold in methylene chloride and then quantified using a GC-FID (Agilent HP-1 column) and identified using a GC-MS (Agilent DB-1 column). Standards were prepared to measure the calibration coefficient for the DPE.

For reactions in flow, DPE was dissolved to 10 wt % (with 1 wt % of *n*-hexadecane used as an internal standard) in *n*-dodecane (89 wt %). The reactor was a 150 mm long and 4.6

mm ID stainless steel HPLC tube (Restek) and placed vertically with flow going from the bottom to the top. The catalyst bed was held in place by quartzwool, and inert rods were used to remove dead volume. The reactor was heated using a 1.5 in. OD aluminum heat sink wrapped in heating tape (Omega, 210 W) connected to a PID controller. The temperature of the catalyst bed was measured directly using a thermocouple touching the bed's downstream side. Gas and liquid products were separated downstream of the back-pressure regulator by using a custom-built separator submerged in an ice bath.

Prior to reaction, the reactor system was purged with He and pressurized with pure H_2 to reaction pressure (5–50 bar) with a back-pressure regulator (Equilibar, U3L series). The ex situ reduced and sieved Ni/SiO_2 was reduced in situ with 20 mL/min of pure H_2 flow at 300 °C for 1 h to remove any passivated NiO . Upon reduction, the reaction temperature (150–300 °C) and gas flow rates (30–225 mL/min) were set. After the reactor was stabilized, the liquid feed was introduced at 200–1600 $\mu\text{L}/\text{min}$ using an HPLC pump (Teledyne SSI, LS-class). Gas products were collected using a 1 L Tedlar gas bag, whereas liquid products were collected in GC vials. Liquid samples were diluted 50-fold in pure methylene chloride for GC analysis, and gas samples were analyzed directly.

The DPE conversion (X_{DPE}) in batch reactions was defined as

$$X_{\text{DPE}} = \frac{N_{\text{DPE}}^i - N_{\text{DPE}}^f}{N_{\text{DPE}}^i} \times 100\% \quad (7)$$

where N_{DPE}^i and N_{DPE}^f are the total moles of DPE at the beginning and end of the reaction, respectively. The DPE residence time (τ) was defined as⁶⁴

$$\tau = t_{\text{rxn}} \times \frac{m_{\text{cat}}}{m_{\text{DPE}}} [\text{min}] \quad (8)$$

where t_{rxn} is the reaction time in minutes while m_{cat} and m_{DPE} are the catalyst and DPE masses, respectively. The DPE conversion in flow reactions was calculated as

$$X_{\text{DPE}} = \sum_j^n \frac{\nu_j C_j}{C_{\text{DPE}}^i} \times 100\% \quad (9)$$

where C_{DPE}^i is the molar concentrations of DPE in the reactor feed, C_j is the concentration of product j in the reactor effluent, and ν_j is the stoichiometric coefficient of product j . The reaction rate ($-r_{\text{DPE}}$) was approximated as

$$-r_{\text{DPE}} = \frac{dC_{\text{DPE}}}{dt} \approx \frac{\Delta C_{\text{DPE}}}{\Delta t} = \frac{\dot{V}_o}{m_{\text{cat}}} \sum_j^n \nu_j C_j [\text{mmol g}_{\text{cat}}^{-1} \text{s}^{-1}] \quad (10)$$

with $\dot{V}_o$ denoting the feed liquid volumetric flow rate to the reactor.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available upon request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.3c05900>.

Polystyrene hydrogenolysis to high-quality lubricants using Ni/SiO_2 (PDF)

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Author Contributions

B.C.V. performed conceptualization, methodology, investigation, formal analysis, and writing—original draft, review, and editing. S.N. developed the methodology and performed investigation, formal analysis, and writing—original draft, review, and editing. D.G.V. performed writing—review and editing, project administration, and funding acquisition.

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Notes

The authors declare no competing financial interest.

ABBREVIATIONS

PS, polystyrene; M_w , average molecular weight; Ni/SiO₂, nickel on silica; PVCH, polyvinyl-cyclohexane; P_H , hydrogen pressure; GCMS, gas chromatography mass spectrometry; GPC, gel-permeation chromatography; MWD, molecular weight distribution; NMR, nuclear magnetic resonance; AAR, alkyl-to-aromatic ratio; VI, viscosity index; DPE, 1,2-diphenylethane; CHBE, 1,2-cyclohexylbenzylethane; DCHE, 1,2-dicyclohexylethane; MCH, methylcyclohexane; TOL, toluene; nC26, *n*-hexacosane; XRF, X-ray fluorescence

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